

Sequences and series - Set 1

TMUA practice questions by topic. Specification MM2.1, MM2.2, MM2.3, M4.19. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.
- A 2440
B 2540
C 1270
D 2600
E 2500
F 2580
- 2 Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$
- A 16
B 6
C 9
D 18
E 24
F 8
- 3 A sequence is defined by $x_{n+1} = 3x_n - 2$ with $x_1 = 2$. Find x_5 .
- A 244
B 80
C 28
D 82
E 162
F 54

- 4 The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.
- A -1650
B -1530
C -1740
D -1590
E -1770
F -1860
- 5 The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.
- A 43
B 189
C 21
D 230
E 39
F 41
- 6 The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.
- A 93
B 186
C 192
D 381
E 126
F 189
- 7 How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?
- A 17
B 16
C 11
D 14
E 18
F 15

- 8** A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.
- A** 30
B 52
C 46
D 94
E 126
F 62
- 9** Which of the following geometric series is/are convergent?
- I** $1 + 0.9 + 0.81 + \dots$
II $3 - 3 + 3 - 3 + \dots$
III $100 + 10 + 1 + 0.1 + \dots$
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III
- 10** An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.
- A** 36
B 32
C 34
D 31
E 33
F 35

- 11** The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?
- A** 21
B 5
C 6
D -5
E 0
F 30
- 12** Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.
- A** 6141
B 3069
C 3072
D 3066
E 6144
F 6138